Lab 10 – Worksheet

Name:	ID:	Section:

Task a. 10-bit counter

Code: Design module `h_counter` & testbench

Provide appropriately commented code for designed module, the code should contain meaningful variable naming. Modify the testbench provided in Fig 10.9(b) to verify the functionality of your designed `h_counter` module.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Results (Waveforms)

**Add snip of relevant signals' waveforms. Make sure the irrelevant area of the snip is cropped.*

Comments

**Observation/Comments on the obtained results/working of code.*

Concept check:

Assuming that a time unit, in above mentioned testbench, is defined in nano seconds; what is the frequency at which the designed `h_counter` module would operate?

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Task b.

Code: Design module v_counter and its testbench

Provide appropriately commented code for designed module and the testbench, the, code should contain meaningful variable naming.

Modify the testbench provided in Error: Reference source not found(b) to verify the functionality of your designed `v_counter` module.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Results (Waveforms)

**Add snip of relevant signals' waveforms. Make sure the irrelevant area of the snip is cropped.*

Comments

**Observation/Comments on the obtained results/working of code.*

Task c. Clock Divider

Before moving to the design module, calculate the div_value using formula provided in manual:

Code: Design module & testbench

Provide appropriately commented code for designed module & its testbench, code should contain meaningful variable naming.
**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Results (Waveforms)

**Add snip of relevant signals' waveforms. Make sure the irrelevant area of the snip is cropped.*

Comments

**Observation/Comments on the obtained results/working of code.*

Exercise

Show your Lab results to the RA and get it signed by the RA.

Calculate the div_value using the formula provided in manual:

Code: Design module & testbench

Provide appropriately commented code for designed module & its testbench, code should contain meaningful variable naming.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

RTL Schematic:

Results (Waveforms)

**Add snip of relevant signals' waveforms. Make sure the irrelevant area of the snip is cropped.*

Comments

*****Observation/Comments on the obtained results/working of code.*

Assessment Rubric
Lab 10 Worksheet
Counter and Clock divider

Name:	Student ID:

Marks Distribution:

		LR2 Code	LR5 Results
In - Lab	Task a	20	10
	Task b	10	10
	Task c	-	-
Exercise		20	20
Total Marks: 100		/50	/40
CLO Mapping		CLO2	CLO2

Affective Domain Rubric		Points	CLO Mapped
AR 7	Viva	/10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	**100**	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/ simulati on model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/ simula tion model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/ simula tion model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/ simulati on /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	**Viva**	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.